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Deep Learning based Object Detection Methods: Current Status and Future Prospects

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***Abstract*—Object Detection has evolved significantly since its inception in the early 2000s when Object Localization was regarded as a challenging task. The transition from Localization to object detection has seen rapid advancements in state-of- the-art techniques. The evolution of object detection can be broadly categorized into Traditional Methods and Deep Learning (DL) Methods. In this paper, we examine both approaches, highlighting the rise of YOLO (You Only Look Once) as a dominant force in Object Detection. We provide a comprehensive survey of all YOLO versions, alongside an overview of pre- YOLO object detection methods. Our discussion also explores optimizing object detection through architectural improvements and data enhancements, while considering future opportunities for increased efficiency.**

***Index Terms*—Deep Learning, Machine Learning, Object Detection, Computer Vision, YOLO.**

I. Introduction

Machine Learning (ML) is found under the umbrella of Artificial Intelligence (AI), and Deep Learning is a subset of ML, hence grouping DL as a branch of Artificial Intelligence [1]. DL connects the dots, much like the human brain, to solve a problem with the help of a logical structure called an Artificial Neural Network [2]. The concept of DL is leading to breakthrough innovations in the fields of computer vision [3], natural language processing [4], strategic games, and many more. Within the expansive realm of AI, this research concentrates on Object Detection. This dynamic field boasts a multitude of applications across various sectors including autonomous vehicles, unmanned aerial systems, demographic analysis, and traffic management.

YOLO (You Only Look Once) is one of various approaches for object detection. The Yolo framework has gained prominence among the numerous methods for object detection due to its amazing speed and accuracy. The realm of object detection is dynamically advancing, with its applications spanning diverse domains such as autonomous vehicles and video surveillance. As technological strides continue to reshape our world, the demand for accurate and efficient algorithms in the field of object detection has intensified. The need for a survey on object detection, particularly focusing on the evolution and prominence of the YOLO. Therefore, this paper sets the stage for an exploration into the evolving landscape of object detection, with YOLO as a central focus.

The remaining sections of this paper are organized as follows: Section 2 discusses the history and background of OD, as well as an in-depth review of existing literature. Section 3 provides a brief description of the datasets and measures utilised for evaluation. Potential research directions are listed in Section 4. Section 5 concludes and summarises the paper.

II. History Of Object Detection

A. Object Detection and its Applications

There can be three objectives in computer vision: Classification, Localization and Detection. Object classification is a method to identify 'what' is present in an image without focusing on the object's exact position. Instead of determining an object's precise location, the focus is on identifying what kind of object it is within the image.

Object localization means not only classifying the object in the image but also finding where the object is in the image [5]. The number of objects within the image distinguishes Localization from OD. Object localization pinpoints the exact position of a single item within an image. In contrast, object detection identifies and locates multiple objects in the same image, as illustrated in Fig.1.



Figure 1. Distinction Among Object Classification, Localization, and Detection

There are many different applications for OD, therefore its possibilities are not limited. The object detection can be applied to detect and classify crops [6]. They can be used in Surveillance and Security [7] in public places like Bus Stand, Railway station, Airport or at Home as well. In Industrial area, they can be used for inspection of defects in

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industrial parts and quality control, which ensures product quality [8]. Talking about sports, Object detection can be used a tool to analyse the details of sports such as freekicks in football, balls crossed the boundaries in cricket, fouls in tennis, if any, and movements of players with summary and insights that can be valuable for coaches, skippers, mentors, and players [9] [10]. In the field of medical, object detection can detect lung disease [11] and Cancer Detection [12]. Crucially, object detection technology is integral to autonomous driving systems. It enables the identification of various entities, including pedestrians [13], cyclists, and other vehicles. There can be many more possibilities for the application of object detection that we have not listed here.

B. Object Detection Methods

Before DL emerged, object detection was a widely used technique. Currently, there are two main approaches to OD: conventional methods and those built upon DL foundations.

1. ***Traditional Detection Methods***: P. Viola et al. introduced a system designed primarily for identifying frontal human faces, but it can also be used to locate other facial elements like eyes, noses, and mouths [14] [15]. The proposed algorithm used weak classification using Haar-like features and Integral Images that allow for very fast feature evaluation. This research also involved developing a classifier using the AdaBoost algorithm. By focusing on a select group of key image features, we created a series of classifiers that progressively concentrate on promising image areas [16]. The training set contains 4916 manually labelled faces with a 24 × 24-pixel resolution.

   The MIT+CMU frontal face Dataset is used for testing [17]. Dalal et al. in 2005 proposed a technique called Histogram of Oriented Gradients in which a detection window slides over an image, and at each location where the window is applied, gradients are calculated [18]. After obtaining the final descriptor, this is fed to the support vector machine, and with the help of “SVM”, classification is done to check if it is the pedestrian or not. The proposed work used two different datasets. The first one is the well-known MIT Pedestrian dataset [19] [20] and the other one is ‘INRIA’ made by them, containing humans cropped from a varied set of 1805 personal images of 64x128 resolution.

2. ***Deep Learning-Based Detection Methods***: OD methods employing DL can be categorized into two primary groups: two-stage and single-stage approaches.

   Two Stage-Detectors: Girshick et al. [21] proposed a Region-based object detection network. It used a ‘Selective Search’ as a region proposal algorithm to propose approximately 2000 regions. These regions are warped first before feeding into the CNN, each warped region is then evaluated by a CNN that performs classification using Linear SVMs and bounding box regression. The Average Precision of 53.7% is achieved on PASCAL VOC Dataset 2010 [22]. Girshick et al. [23] introduced a new approach in 2015 to enhance the RCNN model by processing the entire input image through a convolutional neural network (CNN) backbone to extract features.

   Using these qualities together with a selective search approach, an extensive amount of regions of interest (RoIs) are proposed and passed on to a CNN. In this method, maxpool is applied to the RoI, resulting in a constant-size feature map. Following that, a fully connected neural network uses this feature map to calculate class probabilities and a bounding box. Object detection performance is evaluated using Mean Average Precision (mAP). The model attains an mAP of 66.1% on the PASCAL VOC 2010 benchmark and 66.7% on the PASCAL VOC 2012 benchmark [24]. In another study, S. Ren [25] introduced a new architecture called the Region Proposal Network (RPN), which replaced the conventional ’Selective Search’ method. This innovative approach resulted in a substantial enhancement of the mean Precision score, reaching 70.4% on the benchmark Pascal VOC dataset.

   Single Stage-Detectors: In 2016, J. Redmom concluded that A. Farhadi’s [26] novel approach transformed object detection from a classification to a regression problem. The core concept involves dividing the input image into a textitS x S grid (where S equals 7) and predicting whether each grid cell contains objects. Every grid cell forecasts ’B’ = 2 bounding boxes, each defined by five parameters: (x, y) for the centre relative to the cell, ’w’ for width, ’h’ for height, and a confidence value. Additionally, class probabilities (c) are predicted for each grid cell. The PASCAL VOC 2007 dataset was used for evaluation. With C = 20 classes, the final output is a 7 x 7 x 30 tensor. The model excelled in real-time object detection, attaining a 63.4.

   In 2017, J. Redmon et al. [27] released a new version of Yolo that overcame the shortcomings of Yolo V1. Different ideas have been implemented to improve the architecture such as:

   - All convolutional layers in the architecture employ batch normalization, rendering the dropout layer unnecessary.
   - A High-Resolution Classifier is used in place of the conventional approach of training the network on the entire 224 x 224 resolution. Instead, the training begins with 224 × 224 resolution but fine-tunes the proposed framework for 10 epochs on the ImageNet dataset [28]for a higher resolution of 448 x 448.

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- The fully connected layers are removed, and one maxpoollayer is removed as well to work with higher resolution. The network is designed in such a way that it works on 416 input images rather than 448 x 448. This is doneto focus on a single centre cell. Large objects generally occupy the centre of the image, thus, locating a single spot at the centre rather than many locations is moreefficient. Convolution layers in the architecture wrap the image by a factor of 32, starting with a 416-input image resulting in a 13 x 13 feature map.
- K-means clustering is used with k=5 to find out the trade-off between recall and complexity of the model.
- YOLOv2 adopted the similar naming arrangement as YOLOv1, predicting five values: $_{pa}$, p_b, p_c, p_d, p_e, wherep_e indicates the class probability.
- Features from both high and low-resolution layers are combined using a passthrough layer.
- To make the model robust, it is trained randomly with input sizes ranging from 320 x 320 to 608 x 608 after every 10 batches.

YOLOv2 achieved a mean average precision of 78.6% when evaluated on the PASCAL VOC 2007 benchmark.

In 2018, J. Redmom ended A. Farhadi [29] introduced a new version called YOLOv3 which is made up of a bigger architecture of 53 convolutional layers and this architecture's backbone is called Darknet-53. In addition to predicting the box's dimensions, the model assigns an objectness score of 1 to each bounding box that overlaps most significantly with an actual object in the image compared to all other predicted boxes. In contrast, for class prediction, individual logistic classifiers were employed using binary cross-entropy loss rather than SoftMax. This helped to work on complex problems such as Open Images Dataset [30]. YOLOv3 achieved an average accuracy of 36.2% when evaluated on the COCO dataset [31].

A research team led by A. Bochkovskiy et al. [32] introduced a new iteration of the YOLO object detection system, known as YOLOv4, in the year 2018. The Backbone, Neck, and Head constitute the architecture's primary components. The Backbone is made up of CSPDarknet53 [33], the Neck is made up of SPP [34] and PAN [35] whereas the Head is made up of YOLOv3 architecture. 'Bag-of-Specials' is used to improve the accuracy at the sacrifice of an increase in inference cost. After trials of several architectures for the back- bone, the authors used CSPNet with Mish activation function [36]. 'Bag-of-Freebies' is used in which data augmentation is done. Apart from the normal augmentation like pixel-wise adjustments, the authors used DropBlock [37] methods that worked as a replacement for Dropout. MixUp [38] is used as a Data Augmentation technique in which images are multiplied and superimposed with different coefficient ratios. YOLOv4 achieved a 43.5% average precision score when evaluated on the MS COCO dataset.

G. Jocher [39], the founder and CEO of Ultralytics released YOLOv5, a new iteration of YOLO. This architecture was developed with PyTorch instead of Darknet, and it contained the AutoAnchor Ultralytics algorithm. The Spatial Pyramid Pooling Fast (SPPF) layer is included in the architecture to boost computational efficiency. Batch Normalization is used in each convolutional layer with SiLU activation function [40]. It also used several data augmentation techniques like MixUP, Mosaic, and Copy Paste [41]. YOLOv5 is accessible in five scaled versions, exhibiting an average precision of 55.8%. Chuyi Li et al. [42] released YOLOv6 in 2022. A new backbone is used which is based on RepVGG [43] that works on the principle of parallelism and for the neck, they used PAN with RepBlocks [43]. Task alignment learning technique used in TOOD [44] is utilized in Label assignment in YOLOv6. Classification Varifocal loss [45] along with GIoU [46] regression loss is used with RepOtimizer [47] as a quantization technique that helped for faster detection. This helped in achievement of 52.3% of Average precision.

In 2022, Chien-Yao Wang and colleagues [48] presented YOLOv7, which features an Extended Efficient Layer Aggregation Network (ELAN) [49] architecture. This design improves the model's learning and convergence efficiency by optimizing gradient flow throughout the network.

This architecture applied concatenation-based models to vary model attributes and develop models of different scales in order to satisfy a range of inference speed requirements. This section will detail the specific techniques employed in YOLOv7 to enhance performance without increasing computational cost, often referred to as" bag-of-freebies:

- The planned re-parameterized convolution is used as a technique similar to ResNet [50] residual destruction and DenseNet [51] concatenation. As a result, the identity connection is removed.
- Deep supervision [52] [53] is a common method usedin deep network training. The basic concept is to addan additional auxiliary head to the network's intermediate levels. The parameters of the shallow network are adjusted using an auxiliary loss function, referred to as the" c o a r s e" loss, while the main loss

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function, termedthe" fine" loss, is used to refine the model's overall performance.

- In the convolutional-batch normalization-activation sequence, batch normalization is closely tied to the convolutional layer. When deploying the model, the calculations involved in batch normalization are incorporated into the convolutional layer's parameters, specifically its bias andweights.

- Through a combination of addition and multiplication, implicit information is incorporated with a convolution feature map, which is used by YOLOR [54].
- Mean Teacher employs an Exponential Moving Average (EMA) [55] model, for its final inference stage.

III. Statistical Analysis Of Datasets And Evaluation Metrics

A. Dataset

Multiple data sets have been publicly open for object detection. We will focus on two widely recognized datasets for our analysis: the PASCAL VOC Dataset and the MS COCO Dataset.

1. ***PASCAL VOC Dataset***: PASCAL VOC is a well-known challenge that started in 2005 and ended in 2012 with the death of Mark Everingham [57]. The dataset is categorized into 20 classes, including entities like automobiles, canines, couches, avian life, and humans. The PASCAL VOC 2007 compilation consists of 9,963 pictures with annotations for 24,640 items. Moreover, the PASCAL VOC 2012 collection comprises 11,530 images labelled for 27,450 regions of interest and 6,929 segmentations.

2. ***MS COCO Dataset****:* The MS COCO (Common Objects in Context) dataset is a massive collection of images specifically curated for training computer vision models to identify objects, segment images, locate key points, and generate descriptive text. [58]. The dataset includes 80 classes such as persons, bicycles, vehicles, animals, and numerous objects, is a major benchmark with 330,000 photos, more than 200,000 of which are labelled. Notably, it has surpassed the use of the PASCAL VOC dataset as the dominant dataset since YOLOv3.

B. Metrics for Evaluation

Various metrics are utilized to assess the performance of object detection methods, as outlined below:

1. ***Intersection over Union***: Intersection over Union (IoU) measures the overlap between a predicted bounding box and the ground truth bounding box, as illustrated in Fig. 2. It is calculated by dividing the area of intersection between the two boxes by the total area covered by both boxes combined.

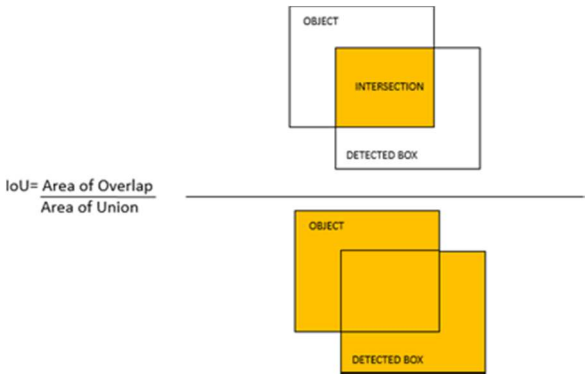


Figure 2. Intersection over Union (IoU)

2. ***Average Precision (AP)***: Before discussing Average Precision, it is crucial to understand Precision and Recall. Precision is determined by dividing the number of correct positive predictions (True Positives) by the total number of positive predictions (True Positives plus False Positives), indicating the accuracy of positive predictions. Recall, on the other hand, measures the proportion of actual positives that were correctly identified, calculated by dividing the number of true positives by the sum of true positives and false negatives. A higher recall may lead to more false positives, reducing precision. Average Precision is employed to balance this trade-off. It considers both Precision and Recall at various confidence thresholds. A higher AP value indicates better model performance. For the PASCAL VOC dataset [59], AP is determined using an 11-point interpolated precision-recall curve. After computing AP for all 20 categories, the overall Average Precision is obtained by averaging the results. In contrast, the COCO dataset uses 101-point interpolation for AP calculation.

C. Summary of Object Detection Methods

Table I provides a comparison of deep learning-based object detection techniques. However, because typical object identification methods are relatively sluggish and already outperform the YOLO architectures, Table II provides a summary of all the YOLO architectures.

TABLE I. Summary Of Deep Learning-Based Object Detection Methods

Model	Proposed Technique	Loss Function
RCNN [21]	Selective Search	Bounding Box regression and Hinge Loss
Fast RCNN [23]	Selective Search	Hinge Loss and Bounding Box regression
Faster RCNN [24]	RPN	Class Log Loss and Bounding Box Regression
YOLO	Classification + Bounding Box Regression	Class sum-squared loss + bounding box regression +object confidence + background confidence

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TABLE II. SUMMARY OF YOLO ARCHITECTURES

Version	Dataset	Average Precision
YOLOv1 [26]	PASCAL VOC 2007	63.4%
YOLOv2 [27]	PASCAL VOC 2007	78.6%
YOLOv3 [29]	COCO 2017	36.2%
YOLOv4 [32]	COCO 2017	43.5%
YOLOv5 [39]	COCO 2017	55.8%
YOLOv6 [42]	COCO 2017	52.3%
YOLOv7 [48]	COCO 2017	56.8%
YOLOv8 [56]	COCO 2017	53.9%

IV. FUTURE RESEARCH DIRECTIONS

Given the advancements in YOLO and its impact on Object Detection, several potential future research directions can be explored:

- ***Enhancing Model Efficiency and Speed***: Focus on developing lightweight architectures and optimizing real-time processing for deployment on edge devices and mobile platforms.
- ***Improving Detection of Small and Occluded Objects***: Advance feature enhancement techniques and novel training strategies to improve YOLO's accuracy in detecting smallor occluded objects.
- ***Expanding Applicability to Multi-Domain and Multi- Modal Detection***: Adapt YOLO for diverse environments, such as underwater or aerial scenarios, and integrate data from multiple sensors to enhance performance in challenging conditions.
- ***Incorporating Explainability and Interpretability***: Developmethods to understand YOLO's decision-making processbetter, improving the transparency and interpretability of the models.
- ***Incorporating and combining multi-features in the YOLOarchitectures***: Techniques from other areas [60] [61] can be employed for better information representations.
- ***Enhancing Robustness Against Adversarial Attacks***: Focus on creating defense mechanisms and employing adversarial training to make YOLO models more resilientin safety-critical applications.
- ***Focusing on Sustainability and Energy Efficiency***: Re- search ways to reduce the energy consumption of YOLO models and optimize them for resource-constrained environments.
- ***Integrating with Autonomous Systems***: Explore the integration of YOLO with autonomous systems like self- driving cars and drones to enhance real-time decision- making capabilities in complex tasks.

These directions provide a roadmap for advancing YOLO and object detection technologies, addressing current limitations, and expanding the application of these models in diverse and emerging fields.

V. CONCLUSION

In this paper, we looked at the evolution of object detection, examining both the early stages characterized by classical approaches and the modern environment dominated by YOLO architectures. In addition, we conducted a study of the datasets and metrics used in evaluating object identification algorithms, giving light on the statistical aspects. Therefore, this study covers the overall evolution of object detection, providing a thorough account of its journey. We anticipate an increase in Average Precision and an increase in processing speed as the YOLO framework evolves. In this paper, we looked at the evolution of object detection, examining both the early stages characterized by classical approaches and the modern environment dominated by YOLO architectures. In addition, we conducted a study of the datasets and metrics used in evaluating object identification algorithms, giving light on the statistical aspects. Therefore, this study covers the overall evolution of object detection, providing a thorough account of its journey. We anticipate an increase in Average Precision and an increase in processing speed as the YOLO framework evolves.

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